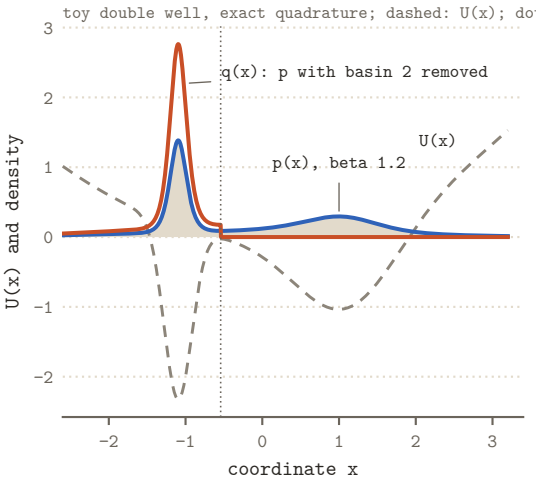
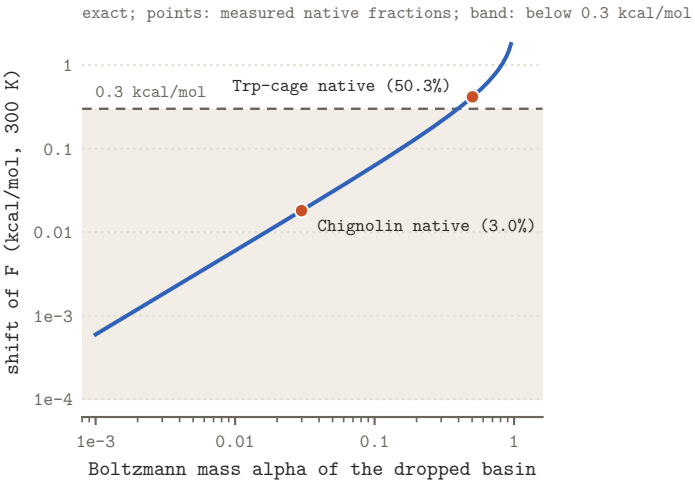

The wide basin holds 50% of the mass
toy double well, exact quadrature; dashed: U(x); dotted: barrier
q(x): p with basin 2 removed
U(x)
p(x), beta 1.2
U(x) and density
coordinate x
3
2
1
0
−1
−2
−2
−1
0
1
2
3


Dropping a basin shifts F by -kT log(1 - alpha)
exact; points: measured native fractions; band: below 0.3 kcal/mol
0.3 kcal/mol
Trp-cage native (50.3%)
Chignolin native (3.0%)
shift of F (kcal/mol, 300 K)
Boltzmann mass alpha of the dropped basin
1
0.1
0.01
1e-3
1e-4
1e-3
0.01
0.1
1